“Talking to AI: What Makes Us Feel Heard?” - A Comparative Study of Emotional Design in Chatbots

Aditi Chaware, Anoushka Dandawate, Neha Chaware, Rucha Limaye, Tanishka Mahajan

Abstract -

As AI chatbots become increasingly integrated into daily life, their ability to provide emotional support is gaining importance particularly among young adults. This study investigates which AI chatbot users prefer in emotionally sensitive conversations and explores the specific conversational design features that shape perceptions of emotional support. We conducted a comparative user study with participants aged 18-25, using two widely available AI chatbots namely ChatGPT and Gemini in emotionally charged scenarios. A mixed-method approach was employed, combining Likert-scale surveys and in-depth semi-structured interviews. Quantitative data were analyzed using Wilcoxon signed rank test to identify significant differences in user ratings, while content analysis was done on qualitative data to extract key features such as empathy, tone, personalization, and validation. A content analysis of open-ended responses further revealed that ChatGPT was more frequently described as empathetic, validating, and emotionally attuned, while Gemini was often perceived as distant or overly directive. These findings highlight the importance of conversational features-such as empathy, tone, and personalization-in shaping user comfort and trust. The study offers practical design implications for creating emotionally intelligent chatbots that users find supportive and human-like.

Keywords- Emotional Connection in Ai chatbot • Empathy in AI • Human-AI interaction • AI conversational features

1. Introduction:

Conversational artificial intelligence (AI) has rapidly evolved in recent years from supporting informational and transactional tasks to offering emotional support, companionship, and even mental health assistance. As these chatbots become more embedded in daily life, the question of how they can cultivate genuine emotional bonds with users has emerged as a focal point in both research and practical design [3] (Chandra et al 2025; Li & Zhang 2024,). The challenge now lies in designing chatbots that go beyond language comprehension to evoke empathy, foster trust, and facilitate meaningful social dialogue, an area that sits at the intersection of technology, psychology, and user experience design. In contrast to traditional human-computer interaction, emotionally intelligent chatbots aim to offer intimacy and psychological reassurance, blurring the boundaries between tool and relational partner [7](Jönsson & Nordberg, 2023).

According to the paper [10] “Why We Tell AI Our Stories: Exploring Motivations, Perceptions and Impact of Interactions with ChatGPT” by Yan Zhuang, users increasingly turn to these systems for affirmation, relief, and companionship, often describing them as “non-judgmental spaces” for self-expression and emotional support . This shift has drawn attention to specific conversational features such as empathetic phrasing, self-disclosure, expressive language, and anthropomorphic cues that influence users’ willingness to trust, disclose, and form attachments to AI agents [8](Papneja and Yadav, 2024; Wei and Li, 2025). Despite their growing popularity, it remains unclear which specific features in chatbot responses actually make users feel heard, understood, or emotionally connected.

Most existing research tends to prioritize functionality and task performance, with limited attention to users' emotional perception and lived experience in support-based contexts. This gap is particularly evident in emotionally sensitive scenarios such as venting, advice-seeking, or coping with stress contexts where emotional resonance may matter more than factual accuracy.
Simultaneously, research underscores both the potential and the paradox of emotionally responsive AI. While such chatbots can enhance satisfaction and emotional engagement, they may also provoke discomfort, confusion, or dependency, especially when users confront their artificial nature or conversational shortcomings [6][5](Li & Zhang, 2024; Chu et al., 2025). These tensions highlight the need to understand how different conversational strategies influence emotional connection across various chatbot designs.

This study addresses these challenges by exploring and comparing how users emotionally perceive the responses of two widely used large language model (LLM) chatbots, ChatGPT and Gemini, in emotionally sensitive interactions. By identifying which design elements (such as tone, empathy, or personalization) contribute most to users' feelings of being emotionally supported, the research aims to inform future design practices for emotionally intelligent conversational agents grounded in human-centered design principles.

To guide this investigation, the study is driven by two key research questions that focus on both user preference and the underlying design choices shaping those preferences.

These questions aim to uncover not just which chatbot is favored, but why users feel more emotionally supported by one over the other, based on specific conversational elements.

- **RQ 1:** Which AI chatbot do users prefer in emotionally sensitive conversations?

- **RQ 2:** What specific conversational design features shape users' perceptions of emotional support in ai chatbots?

2. Literature review:

The importance of emotionally intelligent chatbots has become a focal point in computer science, psychology, and human–AI interaction research, especially for sensitive interactions where emotional nuances are crucial. Recent systematic reviews, such as [2]"Emotionally Intelligent Chatbots: A Systematic Literature Review" (2022), offer a comprehensive overview of over forty studies, highlighting core design strategies like sentiment detection, empathetic phrasing, and conversational flow to enhance emotional support. User perception studies, including [11]"Human–Chatbot Communication: A Systematic Review of Psychological Studies" (2025), further reinforce the significance of empathy and tone in building trust, emotional safety, and comfort with chatbots. [1] Bhandari (2025) outlines an emotion-aware AI chatbot designed to provide immediate mental health support, leveraging Natural Language Processing (NLP) and Machine Learning (ML) to identify user emotions and deliver empathetic responses. This research addresses the critical need for accessible mental health care by proposing a chatbot capable of real-time emotional first aid, aiming to enhance self-awareness and encourage users to seek further professional help when necessary.

While concrete frameworks like the [12] "Empathy-Based Communication Framework for Chatbots" (2023) have emerged focusing on tone calibration, reflective listening, and pacing there remain concerns regarding ethical design, with critiques such as [13] "Emotional Intelligence in AI Chatbots: Advancing Adaptive and Ethical Human–AI Interactions" (2023) warning that over-humanizing chatbots may undermine authenticity and user trust. Technical advancements, exemplified by [14] "Advancing Emotional Intelligence in Chatbots through Deep Learning" (2021), address real-time emotion recognition and adaptive empathy, yet also acknowledge persistent risks like misinterpretation, emotional bias, and inconsistent support. Despite these advances, most existing studies concentrate on individual systems or therapeutic contexts, and there is a notable gap in comparative research involving general-purpose large language models (LLMs) like ChatGPT and Gemini within emotionally sensitive scenarios. Moreover, the influence of specific conversational features such as tone, empathy, encouragement, personalization, or validation on user preference has been insufficiently explored.

To address these gaps, our study employs a mixed-method evaluation of ChatGPT and Gemini, directly examining how design features such as empathy, validation, and tone affect user perception of emotional support and preference. This research aims to deepen understanding of emotionally intelligent chatbot interactions beyond therapeutic contexts, offering practical insights for responsible AI design that balances technical capability, authentic engagement, and ethical considerations

3. Method:

3.1 Study Design:

This research employed a mixed-methods, comparative user study to explore and evaluate how different conversational design features contribute to perceived emotional connection in AI chatbot interactions. We focused specifically on responses from two popular AI chatbots, ChatGPT and Gemini AI, in emotionally sensitive scenarios relevant to young adults.

3.2 Participants:

The participant pool consisted of 20–25 young adults, aged 18 to 25, primarily students and frequent users of AI chatbots. All participants reported regular usage of chatbots like ChatGPT, were comfortable communicating in English, and resided mainly in urban tier-1 cities. Selection emphasized those who engaged in text-based chat and expressed comfort sharing personal perspectives via digital platforms.

3.3 Procedure:

Participants were presented with 2-3 carefully constructed, emotionally sensitive scenarios Each scenario was submitted to both ChatGPT and Gemini AI, generating two distinct chatbot responses for each prompt. Participants reviewed or interacted with both chatbot replies per scenario to ensure a direct, controlled comparison.

Sample scenarios included:

- You didn't do well in an important exam, and you're feeling disappointed and unsure of yourself.
- You had a fight with a close friend, and now you regret how you reacted

3.4 Data Collection:

Data collection followed a sequential, multi-phase process:

1. **Survey Ratings:** After reading each response, participants rated their experience with each chatbot using a 5-point Likert scale. Rating items measured perceived emotional connection, empathy, supportiveness, personalization, and non-judgmental language.
2. **Short Interviews:** Following the completion of all scenarios, participants took part in a brief, semi-structured interview lasting approximately 10 minutes. They reflected on which chatbot response felt more emotionally supportive, reasons for their ratings, and the impact of tone, language, or message structure on their feelings of comfort and trust.
3. **Open-Ended Feedback:** Participants could further elaborate on their preferences and articulate any factors beyond the survey questions that shaped their sense of connection.

All sessions were conducted with strict adherence to ethical protocols. Informed consent was obtained; participants' anonymity and right to withdraw were protected; and emotional well-being was prioritized throughout, as the prompts involved potentially sensitive personal topics.

3.5 Measures:

Quantitative Measures:

A set of 10 Likert-scale items {5 point} per chatbot captured user perceptions of empathy, supportiveness, emotional connection, and personalization.

Qualitative Measures:

Semi-structured interview data and open-ended survey responses were collected to gain nuanced insights into participants' preferences, interpretations, and emotional reactions to chatbot responses.

4. Data Analysis

4.1 Quantitative Analysis:

To examine how users perceive emotional support from two AI chatbots ChatGPT and Gemini within-subjects user study was conducted Each participant interacted with both chatbots across three emotionally sensitive scenarios and rated them on 10 Likert-scale items measuring aspects of emotional connection, such as empathy, tone, validation, and conversational appropriateness.

Participants used a 5-point Likert scale to respond to each item (e.g., "The chatbot acknowledged my feelings," "The chatbot responded in a caring way"). Since each participant rated both chatbots, the resulting data was paired and ordinal in nature.

To compare these paired responses, the Wilcoxon Signed Rank Test, a non-parametric alternative to the paired samples t-test was used. This test is appropriate for ordinal data that does not meet the assumption of normality. It compares the ranks of score differences across conditions to determine if one condition (in this case, one chatbot) consistently received higher or lower scores.

Study Overview

1. **Participants:** Same group for both conditions
2. **Conditions:**
   a. Condition 1: ChatGPT
   b. Condition 2: Gemini
3. **Variables:** 10 emotional support items rated for each chatbot (e.g., "I felt understood by the chatbot")
4. **Scale:** 5-point Likert (ordinal data)
5. **Data type:** Paired responses (within-subject design)

Assumptions

- Participants rated both chatbots on the same questions under similar conditions
- Responses are ordinal and paired
- No assumption of normality required
- Scores are compared within individuals, not across groups

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	Name	String	18	0		None	None	50	Left	Nominal	Input
2	Age	Numeric	8	1		None	None	8	Right	Scale	Input
3	Gender	String	6	0		None	None	50	Left	Nominal	Input
4	Feelings_G...	Numeric	8	1		None	None	8	Right	Scale	Input
5	Friendly_GPT	Numeric	8	1		None	None	8	Right	Scale	Input
6	Val_GPT	Numeric	8	1		None	None	8	Right	Scale	Input
7	Encourage...	Numeric	8	1		None	None	8	Right	Scale	Input
8	Personal_...	Numeric	8	1		None	None	8	Right	Scale	Input
9	Followup_...	Numeric	8	1		None	None	8	Right	Scale	Input
10	Structure_...	Numeric	8	1		None	None	8	Right	Scale	Input
11	Human_GPT	Numeric	8	1		None	None	8	Right	Scale	Input
12	EmoSup_G...	Numeric	8	1		None	None	8	Right	Scale	Input
13	Comf_GPT	Numeric	8	1		None	None	8	Right	Scale	Input
14	Feelings_G...	Numeric	8	1		None	None	8	Right	Scale	Input
15	Friendly_G...	Numeric	8	1		None	None	8	Right	Scale	Input
16	Val_Gem	Numeric	8	1		None	None	8	Right	Scale	Input
17	Encourage...	Numeric	8	1		None	None	8	Right	Scale	Input
18	Personal_...	Numeric	8	1		None	None	8	Right	Scale	Input
19	Followup_...	Numeric	8	1		None	None	8	Right	Scale	Input
20	Structure_...	Numeric	8	1		None	None	8	Right	Scale	Input
21	Human_Gem	Numeric	8	1		None	None	8	Right	Scale	Input
22	EmoSup_G...	Numeric	8	1		None	None	8	Right	Scale	Input
23	Comf_Gem	Numeric	8	1		None	None	8	Right	Scale	Input

Table 1. SPSS Variable View setup for the AI chatbot emotional support study. Each row represents a variable, including participant identifiers (Name, Age, Gender) and paired Likert-scale ratings for ChatGPT and Gemini across ten conversational features. All rating variables are set as scale measures for statistical analysis.

4.2 Qualitative Analysis:

A content analysis was conducted to examine the specific conversational features that participants perceived as emotionally supportive in their interactions with the chatbots. This method enabled systematic identification and quantification of recurring elements in the open-ended responses.

An initial set of 11 predefined codes was developed based on features frequently cited in prior research on emotionally supportive chatbot communication. These codes included:

- Follow-up Questions & Reciprocation – whether follow-up engagement was present
- Response Structure & Length – the conciseness or elaboration of the chatbot's reply
- Encouragement – expressions of motivation, reassurance were offered
- Personalization – responses were tailored to the user's situation
- Empathy – emotional understanding was conveyed
- Tone – the emotional tone of the response
- Validating and Non-Judgmental – the user's feelings were affirmed without criticism
- Feels Personal – the response felt intimate or human-like
- Emojis – emojis to express emotion
- Asking Clarifying Questions - Questions for further understanding
- Offering Suggestions – advice or next steps were provided

During the coding process, additional codes were generated inductively from the data. These emergent codes, such as Relatable & Human, Felt Distant and Detached, Directly Offers Steps Felt Bombarding, and Prefer Human over AI, were added to capture themes that were not anticipated but frequently appeared in participant feedback.

Each response was coded manually, and multiple codes were applied when relevant. Responses were also categorized according to whether they referred specifically to ChatGPT, Gemini, both chatbots, or neither.

Through this approach, patterns in user feedback were identified and interpreted, allowing for insights into which features were most commonly associated with feelings of emotional support during chatbot interactions.

5. Results

5.1 Quantitative results

Descriptive Statistics

Median values show consistently higher ratings for ChatGPT across most items as seen in fig 1

NPar Tests

Descriptive Statistics

	N	Percentiles 25th	50th (Median)	75th
Feelings_GPT	25	4.000	5.000	5.000
Friendly_GPT	25	3.500	4.000	5.000
Val_GPT	25	4.000	4.000	5.000
Encourage_GPT	25	4.000	4.000	5.000
Personal_GPT	25	2.500	3.000	4.000
Followup_GPT	25	3.000	4.000	5.000
Structure_GPT	25	3.000	4.000	5.000
Human_GPT	25	2.500	4.000	5.000
EmoSup_GPT	25	2.000	4.000	5.000
Comf_GPT	25	2.000	4.000	5.000
Feelings_Gem	24	3.000	4.000	4.750
Friendly_Gem	23	2.000	3.000	4.000
Val_Gem	24	3.000	3.500	4.000
Encourage_Gem	24	3.000	3.000	4.000
Personal_Gem	24	2.000	3.000	3.000
Followup_Gem	24	2.000	3.000	4.000
Structure_Gem	24	2.000	3.000	4.000
Human_Gem	24	2.000	3.000	3.000
EmoSup_Gem	24	2.000	3.000	3.750
Comf_Gem	24	1.250	2.500	3.000

Figure 1. Median, 25th, and 75th percentile Likert-scale scores for each conversational feature across ChatGPT and Gemini. ChatGPT consistently received higher median ratings across most dimensions including feelings of being understood, friendliness, validation, encouragement, follow-up, conversational structure, perceived human-likeness, emotional support, and comfort, with the exception of personalisation where the difference was smaller.

Interpretation

The Wilcoxon Signed Rank Test revealed statistically significant differences in 8 out of 10 items, with ChatGPT consistently scoring higher than Gemini. This suggests that participants perceived ChatGPT as more emotionally supportive across most conversational dimensions.

As seen in fig 2, across the ten conversational dimensions assessed, the Wilcoxon Signed Rank Test revealed statistically significant differences in eight, all in favor of ChatGPT. The largest differences were observed for friendliness ($Z = -3.452$, $p < .001$), validation ($Z = -3.218$, $p = .001$), and encouragement ($Z = -3.300$, $p < .001$), suggesting that participants perceived ChatGPT as more socially engaging and emotionally validating compared to Gemini.

Other features such as perceived human-likeness ($Z = -3.240$, $p = .001$), conversational structure ($Z = -3.103$, $p = .002$), and comfort ($Z = -2.996$, $p = .003$) also favored ChatGPT, indicating a consistent advantage in conversational quality and emotional responsiveness.

Two features; personalisation ($Z = -1.975$, $p = .048$) and feeling emotionally supported ($Z = -2.020$, $p = .043$)—showed smaller, less robust differences, suggesting that these aspects may be influenced by factors beyond the immediate conversational interaction. Effect size calculations showed large effects ($r > 0.7$) for most significant results, reinforcing that the differences were not only statistically significant but also practically meaningful.

These findings indicate that ChatGPT's strengths lie in its ability to maintain a warm tone, structure responses effectively, and create a supportive conversational environment, all of which contribute to a stronger perception of emotional connection compared to Gemini.

Wilcoxon Signed Rank Test Results:

Test Statistics[a]

	Feelings_Gem – Feelings_GPT	Friendly_Gem – Friendly_GPT	Val_Gem – Val_GPT	Encourage_Gem – Encourage_GPT	Personal_Gem – Personal_GPT	Followup_Gem – Followup_GPT	Structure_Gem – Structure_GPT	Human_Gem – Human_GPT	EmoSup_Gem – EmoSup_GPT	Comf_Gem – Comf_GPT
Z	-2.884[b]	-3.452[b]	-3.218[b]	-3.300[b]	-1.975[b]	-2.573[b]	-3.103[b]	-3.240[b]	-2.020[b]	-2.996[b]
Asymp. Sig. (2-tailed)	.004	<.001	.001	<.001	.048	.010	.002	.001	.043	.003

a. Wilcoxon Signed Ranks Test

b. Based on positive ranks.

Figure 2. Wilcoxon Signed Rank Test results comparing ChatGPT and Gemini ratings for each conversational feature. Statistically significant differences ($p < .05$) were found for all features, with Z values ranging from -1.975 to -3.452. The negative Z values indicate that ChatGPT scores were generally higher than Gemini scores for the majority of features.

5.2 Qualitative results

A content analysis was conducted on participants' open-ended responses to identify features contributing to their emotional support perceptions. The identified codes, along with their frequencies across ChatGPT, Gemini, generic, and both-agent responses, are presented in Table X.

The most frequently observed code was Empathy (n = 51), followed by Felt Distant and Detached (n = 34) and Emotional Tone (n = 33). ChatGPT responses were predominantly coded as empathetic, emotionally attuned, and validating, whereas Gemini was often perceived as distant or overly directive. Features such as Helpful Suggestions (n = 17), Memory (n = 18), and Personalization (n = 11) were also noted across both agents, reflecting user expectations of intelligent and emotionally responsive behavior.

Additionally, users described ChatGPT as more Relatable & Human (n = 23) and Personal (n = 20).
In contrast, Gemini's responses were more often coded with negative emotional markers.

Participants frequently described Gemini as Felt Distant and Detached (n = 26) and Directly Offers Steps / Felt Bombarding (n = 25).
These findings suggest that emotionally supportive interactions were associated with empathetic tone, personalization, and validation. Emotionally disconnecting interactions were characterized by a lack of warmth, abrupt solution delivery, or robotic tone. Full details of the coded categories and counts are shown in Table 2.

Interpretation

The content analysis revealed that participants perceived ChatGPT as more emotionally supportive, with high frequencies for empathy, emotional tone, and relatability. In contrast, Gemini was often described as distant and overly task-focused.

Features such as helpful suggestions, validation, and personalization were more commonly noted in ChatGPT's responses. The codes and their distribution (see Table 2) suggest that emotional connection was fostered through warmth, empathy, and human-like tone, while a lack of these features led to emotional detachment.

Codes	Count- ChatGpt	Count- Gemini	Generic	Both	Final Count
Follow-up Questions and Reciprocation	15	2	6	2	25
Response structure : concise	12	2	5	-	19
Response structure : lengthy	-	6	-	-	6
Encouragement	3	-	-	1	4
Personalization	4	-	5	2	11
Empathy (Understanding)	37	3	10	1	51
Emotional Tone	23	2	7	1	33
Practical tone (positive)	2	15	2	-	19
Feels personal	18	-	2	-	20
Emojis	1	-	-	-	1
Asking clarifying questions	4	-	1	1	6
Offering Suggestions helpful	4	9	1	3	17
Memory (remembers proir conversations)	15	-	2	1	18
Validating	7	1	2	2	12
Non Judgemental	2	1	1	3	7
Time to express	11	-	2	-	13
Felt distant and detached	2	26	4	2	34
Directly offers steps felt bombarding	-	25	2	1	28
Prefer human over AI	-	-	7	2	9
Relatable & Human	20	1	1	1	23

Table 2: Frequency of qualitative codes describing emotional connection features in AI chatbot interactions. The table shows counts for ChatGPT, Gemini, generic references, both chatbots, and the final total count per code.

6. Findings

ChatGPT was consistently rated higher than Gemini across most emotional support dimensions, with statistically significant differences observed in 8 out of 10 Likert-scale items. Participants found ChatGPT's tone, warmth, validation, and response structure more effective and emotionally supportive. Effect sizes were large, indicating a strong preference for ChatGPT in terms of conversational quality. However, no significant difference was found in perceived personalisation or overall emotional connection, suggesting that while ChatGPT offered a more emotionally intelligent interaction style, the depth of emotional bonding may not have varied as much between the two chatbots.

Content analysis supported these findings, showing that ChatGPT's responses were frequently coded as empathetic, emotionally attuned, and human-like. In contrast, Gemini was often described as distant and overly directive, with fewer instances of empathy or personalization. Overall, emotionally supportive interactions were associated with warmth, validation, and relatability, while emotionally disconnecting ones were linked to impersonal tone and abrupt solution-giving.

7. Implications

The findings highlight the importance of designing emotionally supportive chatbots with anthropomorphic features, such as warmth, empathy, and human-like tone that enhance relatability and foster emotional connection. Participants responded more positively to agents that felt "relatable and human" (n = 23), suggesting that subtle cues like conversational warmth, encouragement, and natural language contribute to perceived emotional intelligence.

Second, participants valued when a chatbot appeared to "know" them through features like memory, contextual awareness, or continuity across conversations. When a chatbot referenced previous interactions or provided personalized feedback, users perceived the experience as more genuine and emotionally engaging. This points to the importance of designing systems that go beyond transactional dialogue and aim to simulate long-term, familiar human relationships.

Another important consideration is the cultural familiarity and popularity of the tool itself. ChatGPT, being widely recognized and associated with AI for many users, may have benefitted from pre-existing expectations of intelligence and helpfulness. This "brand familiarity" effect suggests that user trust and openness can be influenced by a chatbot's broader cultural presence, even before the interaction begins. Designers should acknowledge how expectations shaped by popularity and reputation can affect emotional perception and user engagement. Users placed high value on responses that aligned with their emotional tone. When a chatbot responded too quickly with advice (as noted with Gemini), it was often seen as cold or dismissive. This underscores the importance of emotional pacing, where acknowledgment and validation precede problem-solving.

The ability of a chatbot to personalize interactions either through language, tone, or content was positively associated with emotional support. Users were more likely to feel cared for when the response felt tailored or considerate of their emotional state. Lack of personalization, as seen in Gemini's generic or directive responses, was linked to emotional detachment. The tone and structure of responses played a crucial role. ChatGPT's conversational flow was seen as smoother and more supportive, while Gemini's tone was described as robotic or detached. This suggests that the stylistic quality of interaction, its rhythm, phrasing, and warmth is central to fostering emotional safety.

Lastly, the findings suggest that AI models must be designed with care. If features include emotional memory, enhanced personalization, or tone tuning, users may be more willing to subscribe provided these features meaningfully enhance emotional engagement. However, care should be taken to ensure that basic emotional support features are not entirely paywalled, to preserve accessibility and ethical user care.

Overall, emotionally supportive chatbot design should prioritize empathy, continuity, familiarity, and warmth mirroring the qualities that users seek in human emotional interactions.

Limitations

- Small and homogenous sample: Mostly students, limiting generalizability.
- Predefined emotional scenarios: May not fully reflect real-life emotional complexity.
- Familiarity bias: Participants were more familiar with ChatGPT than Gemini.
- Text-only interaction: Other modalities like voice or visuals were not considered.

8. Conclusion

As AI chatbots become increasingly integrated into emotionally sensitive contexts, understanding how users experience support in these interactions is essential. This study highlights that users' perceptions are not shaped solely by what the chatbot says, but by how it communicates empathy, emotional tone, and a sense of human-like presence deeply influencing trust and comfort. The contrast between ChatGPT and Gemini reveals that even subtle differences in design can lead to meaningful variations in user experience.

These insights emphasize the growing importance of designing emotionally intelligent AI systems. As users begin to form expectations around how these tools should respond in vulnerable moments, emotional intelligence will become as crucial as factual accuracy. Future chatbot development must consider not just linguistic competence but also emotional resonance, continuity, and perceived personality. As we move toward more socially embedded AI tools, designing for emotional support will no longer be optional, it will be essential.

References

[1] Bhandari, H. (n.d.). Emotion-Aware AI Chatbot for Mental Health Support. International Journal of Research Publication and Reviews. Retrieved from https://ijrpr.com/uploads/V6ISSUE4/IJRPR43622.pdf

[2] Bilquise, G., Ibrahim, S., & Shaalan, K. (2022). Emotionally Intelligent Chatbots: A Systematic Literature Review. Semantic Scholar. Retrieved from https://www.semanticscholar.org/paper/Emotionally-Intelligent-Chatbots%3A-A-Systematic-Bilquise-Ibrahim/20c317c78caeaebb2b07cd3334747783faa5bd81

[3] Chandra, M. (2025). Longitudinal Study on Social and Emotional Use of AI Conversational Agent. arXiv. Retrieved from https://arxiv.org/pdf/2504.14112

[4] Chang, Y. (2024). Modeling Emotions and Ethics with Large Language Models. arXiv. Retrievedfrom https://arxiv.org/html/2404.13071v1

[5] Chu, X., Zhang, Y., & Li, J. (2025). Illusions of Intimacy: Emotional Attachment and Emerging Psychological Risks in Human-AI Relationships. arXiv. Retrieved from https://arxiv.org/pdf/2505.11649

[6] Han, L., & Zhang, R. (2024). Finding Love in Algorithms: Deciphering the Emotional Contexts of Close Encounters with AI Chatbots. Journal of Computer-Mediated Communication, 29(5), Article zmae015. Retrieved from https://academic.oup.com/jcmc/article/29/5/zmae015/7742812?login=false

[7] Jönsson, A., & Nordberg, C. (2023). Human-Like Chatbot: A Quantitative Study of the Emotional Response Toward Human-to-Machine Interaction. Retrieved from https://www.diva-portal.org/smash/get/diva2:1761098/FULLTEXT02

[8] Papneja, J., & Yadav, A. (2024). Self-Disclosure to Conversational AI: A Literature Review, Emergent Framework, and Directions for Future Research. Retrieved from https://www.researchgate.net/publication/383605691_Self-disclosure_to_conversational_AI_a_literature_review_emergent_framework_and_directions_for_future_research

[9] Wei, F., Li, Y., & Ding, B. (2025). Towards Anthropomorphic Conversational AI Part I: A Practical Framework. arXiv. Retrieved from https://arxiv.org/abs/2503.04787

[10] Zhuang, Y. (2025). Why We Tell AI Our Stories: Exploring Motivations, Perceptions and Impact of Interactions with ChatGPT. Retrieved from https://www.diva-portal.org/smash/record.jsf?dswid=-6140&pid=diva2%3A1976495&c=1&searchType=SIMPLE&language=en&query=Why+we+tell+ai+our+stories&af=%5B%5D&aq=%5B%5B%5D%5D&aq2=%5B%5B%5D%5D&aqe=%5B%5D&noOfRows=50&sortOrder=author_sort_asc&sortOrder2=title_sort_asc&onlyFullText=false&sf=all

[11] Rafikova, A., & Voronin, A. (2025). Human–chatbot Communication: A Systematic Review of Psychological Studies. AI & Society. https://doi.org/10.1007/s00146-025-02277-y

[12] Lin, S., Lin, L., Hou, C., Chen, B., Li, J., & Ni, S. (2023). Empathy-Based Communication Framework for Chatbots: A Mental Health Chatbot Application and Evaluation. In International Conference on Human-Agent Interaction (HAI '23), ACM, 13 Pages. https://doi.org/10.1145/3623809.3623865

[13] Shukla, P. (2025). Emotional Intelligence in AI Chatbots: Advancing Adaptive and Ethical Human-AI Interactions. Zenodo. https://doi.org/10.5281/zenodo.14838063

[14] Usha Rani, M., Subi Stalin, M., Vinod Kumar P, Dr., Ashok Kumar, Ch., Sandhyarani, M., & Kosuri, U. (2024). Advancing Emotional Intelligence in Chatbots through Deep Learning: A Framework for Real-Time Sentiment and Emotion Recognition. Frontiers in Health Informatics, 13(3), 7880–7897. https://healthinformaticsjournal.com/index.php/IJMI/article/download/676/642/1255